

Supply Chain Executive Dashboard

Report Structure

Project Title: Supply Chain Analysis

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Abstract

Supply chain management generates vast amounts of operational and transactional data across procurement, inventory, logistics, sales, and customer activities. Analyzing this data manually is time-consuming, error-prone, and often fails to provide timely insights for strategic decision-making. This project aims to develop a comprehensive **Supply Chain Executive Dashboard** that transforms raw business data into meaningful, interactive, and actionable insights using **Microsoft Excel, PostgreSQL, Power Query, and Power BI**. The project follows a complete Business Intelligence (BI) workflow, beginning with data collection, cleansing, and transformation in Power Query, followed by database implementation in PostgreSQL. A **Star Schema** data model was designed to optimize analytical performance, while SQL queries were used to extract key business metrics. Power BI was utilized to create interactive dashboards supported by DAX measures and dynamic visualizations. The dashboard provides a centralized view of critical business KPIs, including **Total Sales, Total Profit, Customer Performance, Product Performance, Logistics Efficiency, Order Trends, and Profit Margin**. It also enables users to analyse customer behaviour, identify top-performing products and regions, monitor shipping performance, and detect low-performing products, departments, and geographical locations. By converting complex datasets into easy-to-understand visual reports, this solution helps business executives and managers make faster, data-driven decisions, improve operational efficiency, optimize inventory planning, enhance customer satisfaction, and increase overall business profitability. The project demonstrates the practical application of modern Business Intelligence tools to solve real-world supply chain challenges and support effective organizational decision-making.

Introduction

In today's data-driven business environment, organizations generate large volumes of supply chain data from sales, inventory, customers, and logistics. Analyzing this data manually is time-consuming and often leads to delayed decision-making. Business Intelligence (BI) tools help convert raw data into meaningful insights for better operational and strategic planning. This project develops a Supply Chain Executive Dashboard using Microsoft Excel, Power Query, PostgreSQL, SQL, and Power BI. The data was cleaned, transformed, stored in a PostgreSQL database, and modelled using a Star Schema to support efficient analysis. Interactive dashboards and DAX measures provide real-time insights into sales, profit, customer behaviour, product performance, and logistics. The solution enables decision-makers to monitor key performance indicators (KPIs), identify business trends, and make faster, data-driven decisions through a centralized and interactive reporting platform. Supply chains generate large volumes of transactional data that are difficult to analyze manually. The objective of this project was to transform raw supply chain data into meaningful business insights using Excel, PostgreSQL and Power BI. The dataset was cleaned using Power Query, stored in PostgreSQL, modeled using a Star Schema, and visualized through interactive dashboards. The solution enables executives to monitor sales, profit, customer behaviour, logistics efficiency and low-performing products from a single reporting platform.

Problem Statement

The organization lacked a centralized reporting system to monitor supply chain performance effectively. Sales, customer, product, and logistics data were stored in raw datasets, making analysis slow and inefficient. Manual reporting was time-consuming, prone to errors, and made it difficult to track key performance indicators (KPIs), identify business trends, and support timely, data-driven decision-making. This project addresses these challenges by developing an interactive Business Intelligence dashboard that provides a single, centralized view of supply chain performance.

Objectives

- Develop a centralized Business Intelligence reporting solution for supply chain analysis.
- Design a Star Schema data model to support efficient analytical reporting.
- Perform business analysis using SQL queries and key performance indicators (KPIs).
- Create interactive and dynamic dashboards in Power BI for data visualization.
- Generate actionable insights and recommendations to support data-driven business decisions.

Technology Stack

Tool	Purpose
Microsoft Excel	Data validation, preliminary analysis, and report preparation
Power Query	Data cleaning, transformation, and ETL (Extract, Transform, Load) processes
PostgreSQL	Database creation, table management, SQL querying, and data storage
Power BI	Data modeling, DAX measure creation, interactive dashboards, and data visualization
GitHub	Project documentation, version control, and portfolio sharing

Dataset Description

Column Name	Description
Days for Shipping (Real)	Actual number of days taken to deliver the order.
Days for Shipment (Scheduled)	Planned number of days allocated for shipment.
Benefit per Order	Profit earned from each individual order.
Sales per Customer	Total sales generated by an individual customer.
Delivery Status	Current status of the order delivery (Delivered, Shipping, Late, etc.).
Late Delivery Risk	Indicates whether the order has a risk of late delivery.
Category ID	Unique identifier for the product category.
Category Name	Name of the product category.
Customer City	City where the customer is located.
Customer Country	Country where the customer resides.
Customer First Name	First name of the customer.
Customer ID	Unique identifier for each customer.
Customer Segment	Customer classification (Consumer, Corporate, Home Office, etc.).
Customer State	State or province of the customer.
Department Name	Business department associated with the product.
Order City	City where the order was placed.
Order Country	Country where the order originated.

Column Name	Description
Order Customer ID	Customer ID linked to the order.
Order Date	Date when the order was placed.
Order ID	Unique identifier for each order.
Order Item Card Product ID	Unique identifier for the ordered product.
Order Item Discount	Discount amount applied to the product.
Order Item Discount Rate	Percentage discount applied to the product.
Order Item ID	Unique identifier for each order item.
Order Item Product Price	Selling price of the ordered product.
Order Item Profit Ratio	Profit ratio earned on the product.
Order Item Quantity	Quantity of products ordered.
Sales	Total sales amount generated by the order.
Order Item Total	Final amount after applying discounts.
Order Profit Per Order	Total profit earned from an order.
Order Region	Geographic region where the order was placed.
Order State	State where the order was processed.
Order Status	Current processing status of the order.
Product Name	Name of the ordered product.
Product Price	Original price of the product.
Shipping Date	Date when the order was shipped.
Shipping Mode	Shipping method used (Standard, First Class, Same Day, etc.).

Dataset Summary

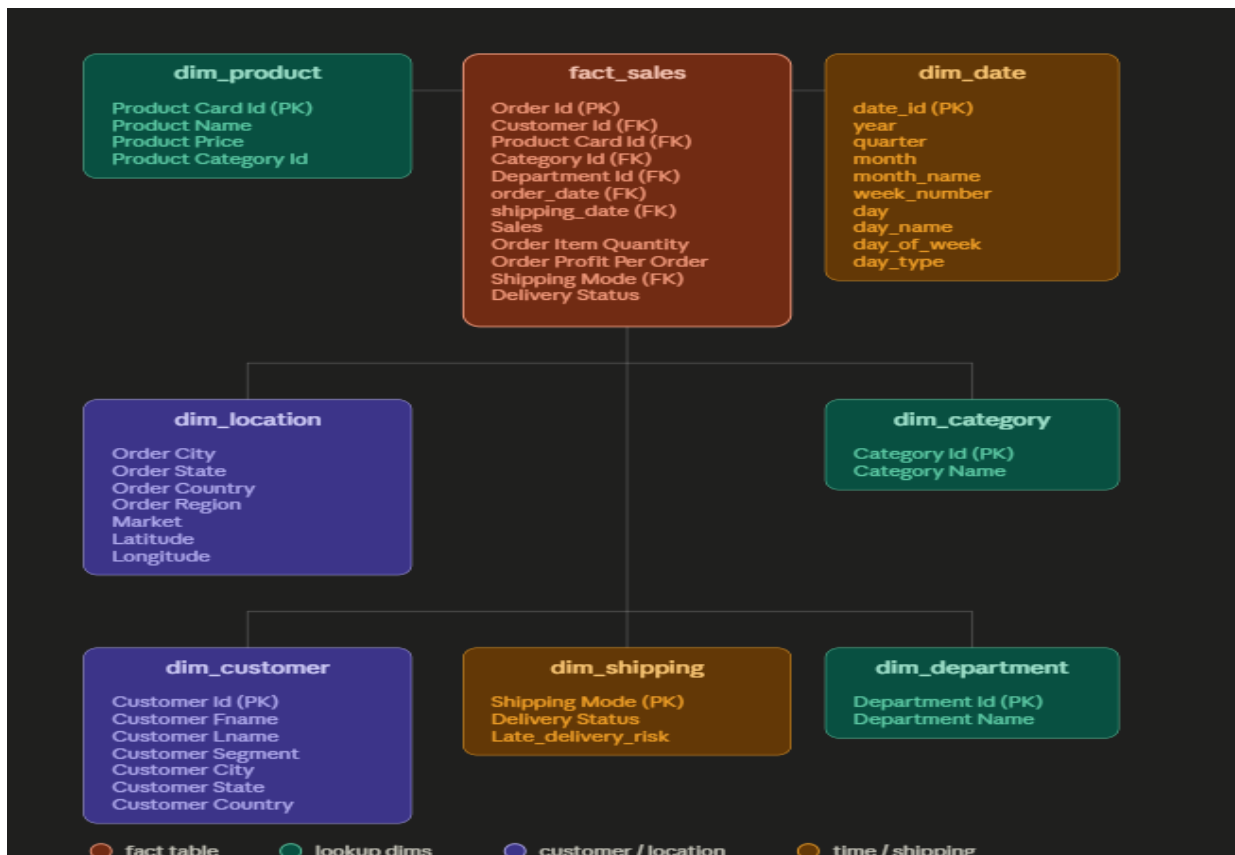
Attribute	Details
Dataset Name	Supply Chain Dataset
Total Records	180,000+
Total Columns	Actual 53 (Used 36)
Data Type	Supply Chain & Logistics Transactions
Source	<u>Kaggle</u>
Database	PostgreSQL
Visualization Tool	Power BI
ETL Tool	Power Query
Reporting Tool	Microsoft Excel

ETL Process

The ETL process began by importing the raw supply chain dataset into Power Query for data preparation and transformation. During the transformation phase, data types were corrected, missing values were handled, date fields were standardized, and column names were organized to improve data consistency and quality. Since the dataset contains valid transactional records, duplicate entries were retained to preserve the integrity of purchase history. The cleaned dataset was then loaded into PostgreSQL, where the database schema and analytical queries were created. Finally, Power BI was connected to the PostgreSQL database to develop a star schema, create DAX measures, and build interactive dashboards for business analysis and decision-making.

Data Model

A **Star Schema** data model was implemented to organize the supply chain data for efficient reporting and analysis. The model consists of a central **Fact_Supply_Chain** table containing transactional data, which is connected to dimension tables including **Customer**, **Product**, **Category**, **Department**, and **Date**. This structure minimizes data redundancy, improves query performance, and enables faster aggregation of key business metrics. It also simplifies Power BI report development by supporting efficient relationships and interactive filtering across multiple dimensions.



Microsoft Excel Executive Report

Objective

The Excel Executive Report provides a summary of key business KPIs, operational performance, top-performing areas, and low-performing areas using PivotTables, formulas, and conditional formatting. It enables management to monitor business performance and make data-driven decisions quickly.

Key Features

- Executive KPI Summary
- Business Performance Summary
- Operations Summary
- Top 3 Performance Analysis
- Low Performance Analysis
- Executive Insights & Management Recommendations

Supply Chain Executive Report						
Total Sales	Average Order Value	Total Profit	Average Profit	Total Customer	Total Order	Profit Margin
36784735	204	3966903	22	20652	65752	10.78
Average Customer Spend	Average Product Discount	Average Product Price	Total Region	Total Order State	Total Country	Average Benefit By Product
183	21	141	23	1089	164	22
Business Performance Summary		Operation Summary				
Category	Fishing	KPI	Value	Status	Business Insight	
Product	Field & Stream Sportsman	Average Shipping Days	3	Good	Delivery Performance Within Target	
Region	Western Europe	Average Late Risk	1	Good	Delivery Process is Stable	
Customer Segement	Consumer	Average Discount	21	Medium	Discount Is Balanced	
Deparment	Fan Shop	Total Quantity	384079	Good	Product Demand Remain Strong	
		Country	164	Good	Global Market Coverage	

EXECUTIVE INSIGHTS

- 1.Fishing category generated the highest sales and profit.
- 2.Consumer segment accounted for the largest share of revenue.
- 3.Western Europe remained the best-performing region.
- 4.Standard shipping maintained delivery within target.

MANAGEMENT RECOMMENDATIONS

- Increase inventory for high-demand products.
- Reduce discounts on low-margin products.
- Improve marketing for slow-moving products.
- Review supplier performance in underperforming regions.

Top 3 Performance

Top 3 Products

Product Name	Sales	Profit	Margin
Diamondback Women's Serene Classic Comfort Bi	4118426	756221	18.4
Field & Stream Sportsman 16 Gun Fire Safe	6929654	493828	7.1
Perfect Fitness Perfect Rip Deck	4421143	427456	9.7

Top 3 Customer

Customer Name	Number of Orders	Sales	Profit	Margin
Mary	21771	12986525	1418869	10.9
James	616	363024	43757	12.1
Robert	577	353734	41441	11.7

Top 3 State

State Name	Number of Orders	Sales	Profit	Margin
California	2566	1485582	1321736	89.0
New York	1692	1013284	843810	83.3
Florida	1653	984827	494826	50.2

Top 3 Department

Department Name	Number of Orders	Sales	Profit	Margin
Electronics	41303	17113871	1834155	10.7
Apparel	35190	7976255	881883	11.1
Home	25889	4609028	497524	10.8

Top 3 Month

Month Name	Number of Orders	Sales	Profit	Margin
Jan	7427	3459246	367127	10.6
Aug	5296	3187032	360210	11.3

Low Performance

Least Performing Bottom Products

Product Name	Number of Orders	Sales	Reason
CDs of rock	271	3060	Low customer demand and low sales volume.
Polar Loop Activity Tracker	53	5937	Limited order frequency generation.
Bowflex SelectTech 1090 Dumbbells	10	6000	Low order quantity despite premium pricing.

Customer Name

Name	Number of Orders	Sales	Reason
Asley	1	11	Very low purchase frequency and revenue contribution.
Brees	1	11	Minimal order activity during the reporting period.
Lynna	1	12	Low sales value and limited purchasing behavior.

State

State Name	Number of Orders	Sales	Reason
Mashonalandia Occidental	1	80	Low demand and limited customer base.
Serbia Central	1	120	Low demand and low order count.
Suceava	1	100	Low sales volume with fewer completed orders.

Department

Name	Number of Orders	Sales	Reason
Book Shop	405	12587	Lower profit margin despite moderate sales.
Health and Beauty	362	106080	Low revenue generation and weak product demand.
Pet Shop	492	41525	Low sales volume and below-target profitability.

Month

Month Name	Number of Orders	Sales	Reason
Dec	5692	2599160	Low Performance Month Behind the Reason should be Seasonal Sales And Other Reason May be High Discount
Nov	5505	2703625	

SQL Implementation

More than 45 SQL queries were written to calculate total sales, profit, monthly trends, regional performance, top customers, product profitability, shipping analysis and customer segmentation. Aggregate functions, CASE statements, GROUP BY, JOINS and window functions were used.

--1 Total Sales

```
SELECT SUM("Sales") AS total_sales FROM fact_sales;
```

	total_sales numeric
1	36784734.31

--2 Total Profit

```
SELECT SUM("Order Profit Per Order") AS total_profit FROM fact_sales;
```

	total_profit numeric
1	3966902.97

--3 Total Orders

```
SELECT COUNT(DISTINCT "Order Id") FROM fact_sales;
```

	count bigint
1	65752

--4 Total Customers

```
SELECT COUNT(DISTINCT "Customer Id") FROM fact_sales;
```

	count bigint
1	20652

--5 Total Quantity

```
SELECT SUM("Order Item Quantity") FROM fact_sales;
```

	sum bigint
1	384079

--6 Average Sales

```
SELECT ROUND(AVG("Sales"),2) FROM fact_sales;
```

	round numeric
1	203.77

--7 Average Profit

```
SELECT ROUND(AVG("Order Profit Per Order"),2) FROM fact_sales;
```

	round numeric
1	21.97

--8 Profit Margin

```
SELECT ROUND(SUM("Order Profit Per Order")*100/SUM("Sales"),2) FROM fact_sales;
```

	round numeric
1	10.78

--9 Monthly Sales

```
SELECT DATE_TRUNC('month',"order_date"),SUM("Sales")
```

```
FROM fact_sales
```

```
GROUP BY 1 ORDER BY 1;
```

	date_trunc timestamp without time zone	sum numeric
1	2015-01-01 00:00:00	1051590.06
2	2015-02-01 00:00:00	927009.88
3	2015-03-01 00:00:00	1051253.67
4	2015-04-01 00:00:00	1014463.26
5	2015-05-01 00:00:00	1050478.42
6	2015-06-01 00:00:00	1024006.15
7	2015-07-01 00:00:00	1038081.17
8	2015-08-01 00:00:00	1029494.67
9	2015-09-01 00:00:00	1018338.58
10	2015-10-01 00:00:00	1049154.25
11	2015-11-01 00:00:00	1029120.22
12	2015-12-01 00:00:00	1057840.86
13	2016-01-01 00:00:00	1046308.23
14	2016-02-01 00:00:00	968542.83
15	2016-03-01 00:00:00	1025853.10
16	2016-04-01 00:00:00	1001211.56
17	2016-05-01 00:00:00	1029400.18

--10 Monthly Profit

```
SELECT DATE_TRUNC('month',"order_date"),SUM("Order Profit Per Order")
```

```
FROM fact_sales
```

```
GROUP BY 1 ORDER BY 1;
```

	date_trunc timestamp without time zone 🔒	sum numeric 🔒
1	2015-01-01 00:00:00	111660.74
2	2015-02-01 00:00:00	99140.66
3	2015-03-01 00:00:00	113778.21
4	2015-04-01 00:00:00	108083.68
5	2015-05-01 00:00:00	112147.90
6	2015-06-01 00:00:00	110147.16
7	2015-07-01 00:00:00	115624.06
8	2015-08-01 00:00:00	117979.77
9	2015-09-01 00:00:00	113467.94
10	2015-10-01 00:00:00	101757.87
11	2015-11-01 00:00:00	104427.36
12	2015-12-01 00:00:00	110641.55
13	2016-01-01 00:00:00	106780.95
14	2016-02-01 00:00:00	86809.49
15	2016-03-01 00:00:00	101259.99

--11 Yearly Sales

```
SELECT EXTRACT(YEAR FROM "order_date"),SUM("Sales")
```

```
FROM fact_sales
```

```
GROUP BY 1 ORDER BY 1;
```

	extract numeric 🔒	sum numeric 🔒
1	2015	12340831.1
2	2016	12303817.0
3	2017	11808435.9
4	2018	331650.1

--12 Top Customers

```
SELECT "Customer Id",SUM("Sales")  
  
FROM fact_sales  
  
GROUP BY 1  
  
ORDER BY 2 DESC  
  
LIMIT 10;
```

	Customer Id  integer	sum  numeric
1	791	10524.17
2	9371	9299.03
3	8766	9296.14
4	1657	9223.71
5	2641	9130.92
6	1288	9019.11
7	3710	9019.10
8	4249	8918.85
9	5654	8904.95
10	5624	8761.98

--13 Repeat Customers

```
SELECT "Customer Id",COUNT(*)  
  
FROM fact_sales  
  
GROUP BY 1  
  
HAVING COUNT(*)>1  
  
Limit 10;
```

	Customer Id  integer	count  bigint
1	11233	17
2	4790	8
3	3936	14
4	5468	11
5	4326	16
6	2520	15
7	2466	21
8	9038	18
9	5697	2
10	1750	17
11	4321	10
12	11092	11
13	176	12
14	576	16
15	9719	6

--14 Customer Lifetime Value

```
SELECT "Customer Id",SUM("Sales") CLV
```

```
FROM fact_sales
```

```
GROUP BY 1
```

```
ORDER BY CLV DESC;
```

	Customer Id  integer	clv  numeric
1	791	10524.17
2	9371	9299.03
3	8766	9296.14
4	1657	9223.71
5	2641	9130.92
6	1288	9019.11
7	3710	9019.10
8	4249	8918.85
9	5654	8904.95
10	5624	8761.98
11	5715	8595.13
12	1464	8409.22
13	10235	8404.22
14	664	8394.26
15	10351	8339.26

--15 Top Products

```
SELECT p."Product Name",SUM(f."Sales")
```

```
FROM fact_sales f

JOIN dim_product p

ON f."Product Card Id"=p."Product Card Id"

GROUP BY 1

ORDER BY 2 DESC

LIMIT 10;
```

--16 Bottom Products

```
SELECT p."Product Name",SUM(f."Sales")

FROM fact_sales f

JOIN dim_product p

ON f."Product Card Id"=p."Product Card Id"

GROUP BY 1

ORDER BY 2

LIMIT 10;
```

--17 Category Sales

```
SELECT c."Category Name",SUM(f."Sales")

FROM fact_sales f

JOIN dim_category c

ON f."Category Id"=c."Category Id"

GROUP BY 1

ORDER BY 2 DESC;
```

--18 Category Profit

```
SELECT c."Category Name",SUM(f."Order Profit Per Order")

FROM fact_sales f

JOIN dim_category c
```



```
ON f."Category Id"=c."Category Id"
```

```
GROUP BY 1;
```

--19 Department Sales

```
SELECT d."Department Name",SUM(f."Sales")
```

```
FROM fact_sales f
```

```
JOIN dim_department d
```

```
ON f."Department Id"=d."Department Id"
```

```
GROUP BY 1;
```

--20 Shipping Mode Sales

```
SELECT "Shipping Mode",SUM("Sales")
```

	Shipping Mode character varying (50) 🔒	sum numeric 🔒
1	First Class	5674369.65
2	Same Day	1942528.52
3	Second Class	7145444.68
4	Standard Class	22022391.46

```
FROM fact_sales
```

```
GROUP BY 1;
```

--21 Delivery Status Count

```
SELECT "Delivery Status",COUNT(*)
```

```
FROM fact_sales
```

```
GROUP BY 1;
```

	Delivery Status character varying (50) 🔒	count bigint 🔒
1	Advance shipping	41592
2	Late delivery	98977
3	Shipping canceled	7754
4	Shipping on time	32196

--22 Average Shipping Days

```
SELECT AVG("shipping_date"-"order_date")
```

FROM fact_sales;

	avg interval 	
1	3 days 11:58:18.5325...	

--23 Delayed Orders (>5 Days)

SELECT COUNT(*)

FROM fact_sales

WHERE "shipping_date"-"order_date">5;

--24 Running Sales

SELECT "order_date",

SUM("Sales") OVER(ORDER BY "order_date")

FROM fact_sales;

	order_date timestamp without time zone 	sum numeric 
1	2015-01-01 00:00:00	299.98
2	2015-01-01 00:21:00	879.96
3	2015-01-01 00:21:00	879.96
4	2015-01-01 00:21:00	879.96
5	2015-01-01 01:03:00	1579.81
6	2015-01-01 01:03:00	1579.81
7	2015-01-01 01:03:00	1579.81
8	2015-01-01 01:03:00	1579.81
9	2015-01-01 01:24:00	2709.67
10	2015-01-01 01:24:00	2709.67
11	2015-01-01 01:24:00	2709.67

--25 Running Profit

SELECT "order_date",

SUM("Order Profit Per Order") OVER(ORDER BY "order_date")

FROM fact_sales;

--26 Rank Customers

```
SELECT "Customer Id",  
  
SUM("Sales"),  
  
RANK() OVER(ORDER BY SUM("Sales") DESC)  
  
FROM fact_sales  
  
GROUP BY 1;
```

	Customer Id integer 	sum numeric 	rank bigint 
1	791	10524.17	1
2	9371	9299.03	2
3	8766	9296.14	3
4	1657	9223.71	4
5	2641	9130.92	5
6	1288	9019.11	6
7	3710	9019.10	7
8	4249	8918.85	8
9	5654	8904.95	9
10	5624	8761.98	10

--27 Dense Rank Products

```
SELECT p."Product Name",  
  
SUM(f."Sales"),  
  
DENSE_RANK() OVER(ORDER BY SUM(f."Sales") DESC)  
  
FROM fact_sales f  
  
JOIN dim_product p  
  
ON f."Product Card Id"=p."Product Card Id"  
  
GROUP BY 1;
```

--28 Row Number Orders

```
SELECT "Order Id",  
  
ROW_NUMBER() OVER(ORDER BY "Sales" DESC)  
  
FROM fact_sales;
```

	Order Id integer 	row_number bigint 
1	68703	1
2	68821	2
3	68859	3
4	68722	4
5	68809	5
6	68724	6
7	68858	7
8	68736	8
9	68875	9
10	68837	10
11	68883	11
12	68766	12
13	68848	13
14	68778	14

--29 NTILE Customer Segments

```
SELECT "Customer Id",  
  
SUM("Sales"),  
  
NTILE(4) OVER(ORDER BY SUM("Sales") DESC)  
  
FROM fact_sales  
  
GROUP BY 1;
```

	Customer Id integer	sum numeric	ntile integer
1	791	10524.17	1
2	9371	9299.03	1
3	8766	9296.14	1
4	1657	9223.71	1
5	2641	9130.92	1
6	1288	9019.11	1
7	3710	9019.10	1
8	4249	8918.85	1
9	5654	8904.95	1
10	5624	8761.98	1
11	5715	8595.13	1
12	1464	8409.22	1
13	10235	8404.22	1
14	664	8394.26	1
15	10351	8339.26	1

--30 LAG Sales

```
SELECT "order_date","Sales",
LAG("Sales") OVER(ORDER BY "order_date")
FROM fact_sales;
```

--31 LEAD Sales

```
SELECT "order_date","Sales",
LEAD("Sales") OVER(ORDER BY "order_date")
FROM fact_sales;
```

--32 Monthly Growth

```
WITH m AS(
SELECT DATE_TRUNC('month',"order_date") dt,
SUM("Sales") sales
FROM fact_sales
GROUP BY 1)
```

```
SELECT dt,sales,  
sales-LAG(sales) OVER(ORDER BY dt)  
FROM m;
```

--33 Highest Sales Order

```
SELECT *  
FROM fact_sales  
ORDER BY "Sales" DESC  
LIMIT 1;
```

--34 Lowest Sales Order

```
SELECT *  
FROM fact_sales  
ORDER BY "Sales"  
LIMIT 1;
```

--35 Negative Profit Orders

```
SELECT *  
FROM fact_sales  
WHERE "Order Profit Per Order"<0;
```

--36 High Value Orders

```
SELECT *  
FROM fact_sales  
WHERE "Sales">  
(SELECT AVG("Sales") FROM fact_sales);
```

--37 Sales Contribution by Category

```
SELECT c."Category Name",  
ROUND(SUM(f."Sales")*100/
```

```
(SELECT SUM("Sales") FROM fact_sales),2)

FROM fact_sales f

JOIN dim_category c

ON f."Category Id"=c."Category Id"

GROUP BY 1;
```

--38 Average Product Price

```
SELECT AVG("Product Price")

FROM dim_product;
```

--39 Most Expensive Products

```
SELECT "Product Name","Product Price"

FROM dim_product

ORDER BY "Product Price" DESC

LIMIT 10;
```

--40 Cheapest Products

```
SELECT "Product Name","Product Price"

FROM dim_product

ORDER BY "Product Price"

LIMIT 10;
```

--41 Product Count by Category

```
SELECT "Product Category Id",COUNT(*)

FROM dim_product

GROUP BY 1;
```

--42 Customer Segment Count

```
SELECT "Customer Segment",COUNT(*)

FROM dim_customer

GROUP BY 1;
```

--43 Department Count

```
SELECT "Department Name",COUNT(*)  
  
FROM dim_department  
  
GROUP BY 1;
```

--44 Best Shipping Mode

```
SELECT "Shipping Mode",  
  
AVG("Order Profit Per Order")  
  
FROM fact_sales  
  
GROUP BY 1  
  
ORDER BY 2 DESC;
```

--45 Top Product in Each Category

```
SELECT *  
  
FROM(  
  
SELECT c."Category Name",  
  
p."Product Name",  
  
SUM(f."Sales") sales,  
  
RANK() OVER(  
  
PARTITION BY c."Category Name"  
  
ORDER BY SUM(f."Sales") DESC) rnk  
  
FROM fact_sales f  
  
JOIN dim_product p  
  
ON f."Product Card Id"=p."Product Card Id"  
  
JOIN dim_category c  
  
ON f."Category Id"=c."Category Id"  
  
GROUP BY c."Category Name",p."Product Name") x  
  
WHERE rnk=1;
```


Dashboard Analysis

1.Supply Chain Executive



Provides a high-level overview of overall supply chain performance including sales, orders, profitability, customers, and regional performance.

KPI	Logic	Business Meaning
Total Sales	Sum of Sales	Total revenue generated from all orders.
Total Orders	Count of Order ID	Number of customer orders received.
Total Profit	Sum of Order Profit	Overall business profitability.
Total Customers	Distinct Customer ID	Total unique customers served.
Average Order Value	Total Sales ÷ Total Orders	Average revenue generated per order.
Profit Margin	Total Profit ÷ Total Sales ×100	Percentage of sales converted into profit.

➤ **Business Objective**

Provide a high-level overview of overall sales, profitability, customer performance, and regional business performance.

➤ **Key Insights**

Overall Business Performance

- Total Sales reached 37Million, indicating strong business growth.
- The company processed 66K customer orders, showing a high transaction volume.
- Overall Profit is 4Million, resulting in a healthy business performance.
- 21K unique customers purchased products.
- Average Order Value is 559, indicating stable customer spending.
- Profit Margin is 10.78%, showing profitable operations.

➤ **Monthly Sales Trend**

Insight

- Sales remained relatively stable from January to October.
- Sales declined noticeably during November and December.

Business Logic

- The decline may be due to seasonal demand reduction or inventory shortages.
- Management should introduce promotional campaigns during low-sales months.

➤ **Top Performing Categories**

Fishing generated the highest sales, followed by:

- Cleats
- Camping & Hiking
- Cardio Equipment
- Women's Apparel

Business Logic

- These categories contribute the largest share of company revenue.
- Inventory and marketing budgets should prioritize these categories.

➤ **Top Performing Regions**

Highest revenue comes from:

- Western Europe
- Central America
- South America

Business Logic

- These regions have the strongest customer demand.
- Expanding logistics and marketing in these regions can further increase sales.

➤ **Executive Recommendation**

- Focus marketing efforts on high-performing categories.
- Improve year-end sales through promotional campaigns.
- Expand operations in top-performing regions.
- Monitor low-performing months to reduce revenue fluctuations.

2.Customer Dashboard



➤ Business Objective

Analyzed customer purchasing behaviour, customer segmentation, and geographical distribution. Customer Segmentation: Consumer Segment contributes approximately 52% of total sales, making it the largest customer group. Corporate customers contribute around 30%. Home Office customers contribute approximately 18%.

Business Logic

- Consumer customers are the company's primary revenue source.
- Customer retention programs should mainly target Consumer customers.

➤ Top Customers

One customer ("Mary") generated significantly higher sales than other customers.

Business Logic

- The business depends heavily on a few high-value customers.
- Customer loyalty programs should be implemented to retain these valuable customers.

➤ Customer Geography

Highest sales come from:

- European Union
- Puerto Rico

Business Logic

- These markets generate strong customer demand.
- Future expansion should prioritize these locations.

➤ **Monthly Customer Trend**

Customer sales remain stable throughout most of the year before declining in the last quarter.

Business Logic

- Customer demand weakens during year-end.
- Seasonal campaigns can improve customer engagement.

➤ **Average Sales per Customer**

Home Office customers have the highest average purchase value.

Corporate customers rank second.

Consumer customers have the lowest average order value.

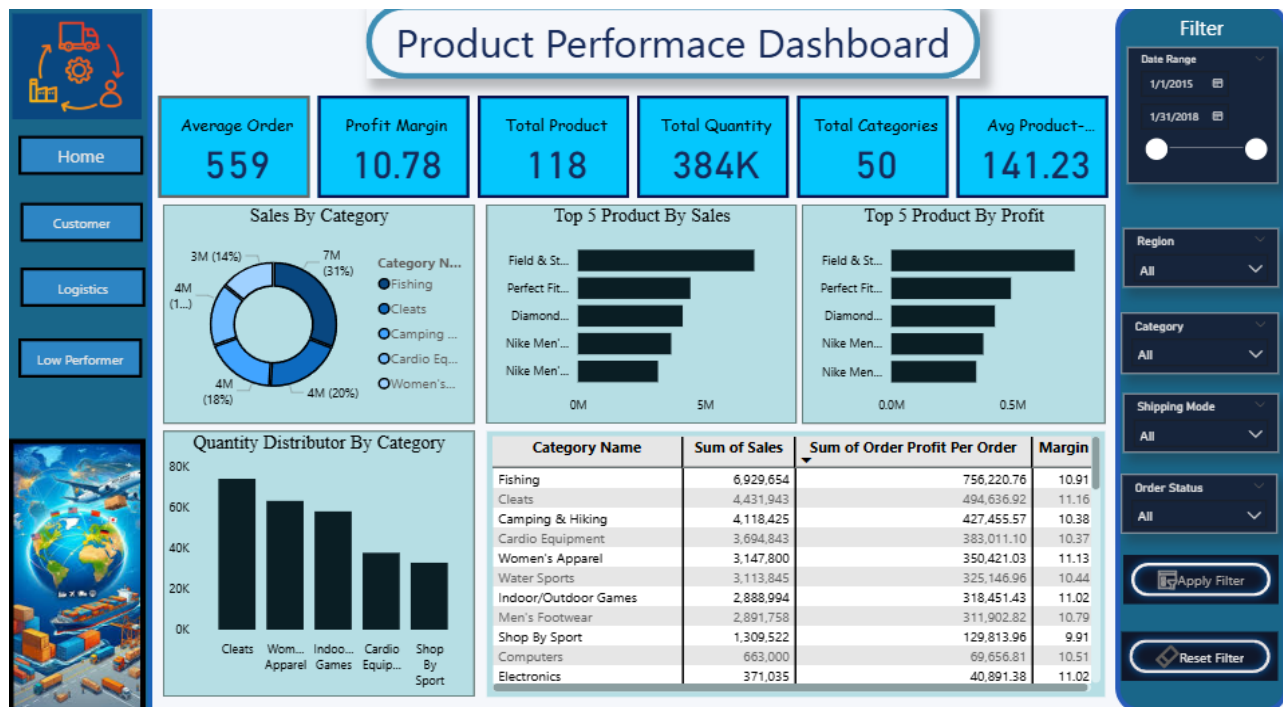
Business Logic

- Although Consumers generate the most revenue, Home Office customers spend more per order.
- Premium products can be targeted toward Home Office customers.

➤ **Executive Recommendation**

- Retain high-value customers with loyalty benefits.
- Increase marketing toward Home Office customers.
- Expand customer acquisition in strong-performing countries.

3.Product Performance Dashboard



➤ Business Objective

Evaluate product sales, profitability, category performance, and inventory demand.

Product Overview

- Total Products: **118**
- Total Categories: **50**
- Total Quantity Sold: **384K**
- Average Product Sales: **141**

➤ Category Performance

Fishing is the highest-selling category. Other strong categories include:

- Cleats
- Camping & Hiking
- Cardio Equipment

Business Logic

- These products generate the majority of company revenue.
- Inventory replenishment should prioritize these categories.

➤ Best Selling Products

Top-selling products belong mainly to:

- Field & Stream
- Perfect Fitness
- Diamondback
- Nike

Business Logic

- These products should receive higher inventory allocation.
- Stock shortages should be avoided.

➤ **Highest Profit Products**

The same products generating high sales also generate high profit.

➤ **Business Logic**

- Strong sales directly contribute to company profitability.
- Continue investing in these profitable products.

➤ **Quantity Distribution**

Cleats have the highest quantity sold.

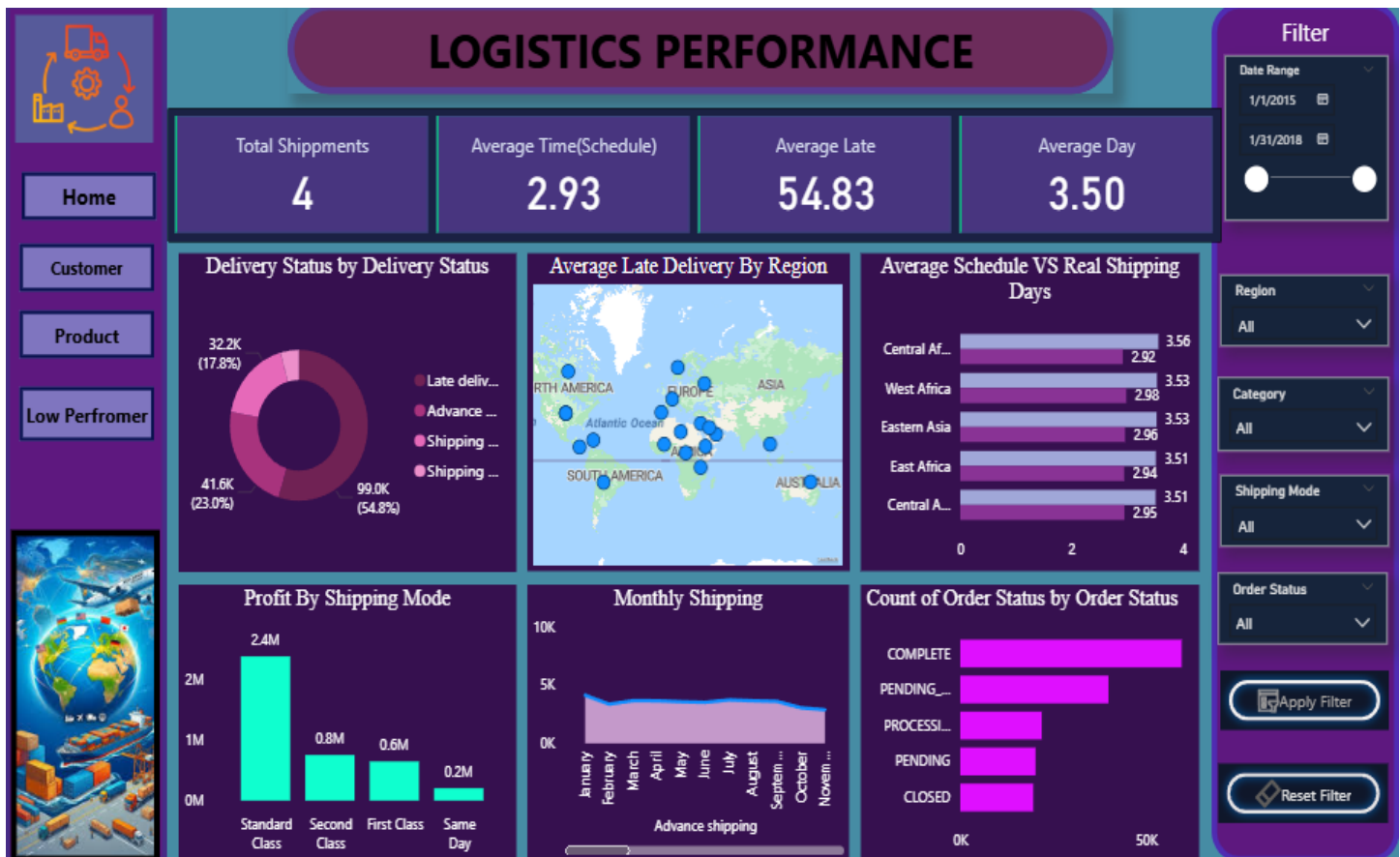
Business Logic

- High sales volume indicates consistent customer demand.
- Forecast inventory carefully to prevent stock-outs.

Executive Recommendation

- Increase inventory for top-selling products.
- Promote profitable products through targeted marketing.
- Monitor low-demand products for possible discontinuation.

4. Logistics Performance Dashboard



➤ Business Objective

Evaluate shipping efficiency, delivery performance, and logistics operations.

Delivery Status

Most deliveries were completed successfully.

Only a small percentage experienced delays.

Business Logic

- The logistics network performs efficiently.
- Delayed deliveries should be investigated to improve customer satisfaction.

➤ Late Delivery by Region

Some regions experience consistently higher delivery delays.

Business Logic

- Logistics resources should be optimized in these regions.

- Route planning should be reviewed.

>Shipping Mode Profitability

Standard Shipping generates the highest profit.

Second Class and First Class contribute moderate profit.

Same Day Shipping contributes the lowest profit.

Business Logic

- Standard Shipping provides the best balance between cost and profitability.
- Same Day Shipping should be reviewed due to higher operational costs.

➤ Monthly Shipping Trend

Shipping volume remains relatively stable throughout the year.

Business Logic

- Logistics operations are well-balanced.
- Capacity planning appears effective.

➤ Order Status

Most orders are completed successfully with very few pending or cancelled orders.

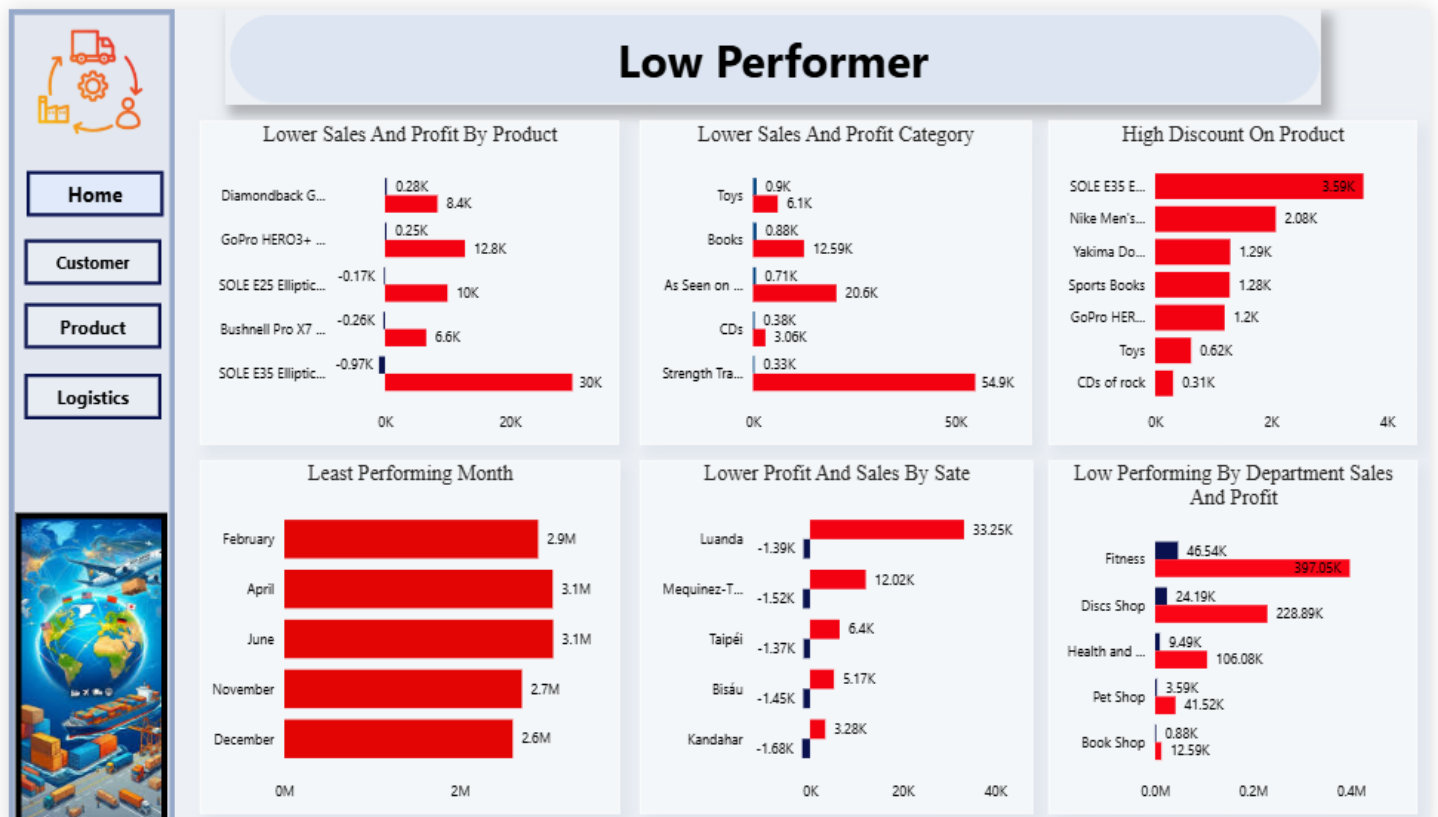
Business Logic

- Order fulfilment efficiency is high.
- Continue monitoring pending orders to reduce delays.

➤ Executive Recommendation

- Improve delivery performance in high-delay regions.
- Promote Standard Shipping where appropriate.
- Reduce late deliveries through better route optimization.

5.Low Performer Dashboard



➤ Business Objective

Identify products, categories, regions, and departments that negatively impact business performance.

➤ Low Performing Products

Several products generate high sales but very low or negative profit.

Business Logic

- Heavy discounts or high operational costs reduce profitability.
- Pricing strategies should be reviewed.

➤ Low Performing Categories

Categories such as:

- Toys
- Books
- CDs

generate weak profit despite reasonable sales.

Business Logic

- These categories require pricing optimization or cost reduction.

➤ **High Discount Products**

Some products receive extremely high discounts.

Business Logic

- Excessive discounting directly reduces company profit.
- Discount policies should be revised.

➤ **Weak Sales Months**

February and November show the weakest sales performance.

Business Logic

- Seasonal promotions should be introduced during these months.

➤ **Low Profit Regions**

Certain cities consistently report negative profit.

Business Logic

- High transportation costs or weak market demand may be reducing profitability.
- Regional operations should be evaluated.

➤ **Low Performing Departments**

Some departments generate high sales but relatively low profit.

Business Logic

- Revenue alone does not indicate business success.
- Profitability should be the primary performance metric.

➤ Overall Business Conclusion

- The company demonstrates **strong overall financial performance**, with **37M in sales**, **4M in profit**, and a **10.78% profit margin**.
- **Fishing, Cleats, and Camping & Hiking** are the strongest product categories.
- **Western Europe and Central America** are the highest-performing regions.
- **Consumer customers** contribute the highest revenue, while **Home Office customers** spend the most per order.
- **Standard Shipping** is the most profitable logistics option.
- Several products and categories experience **high discounts and low profitability**, indicating opportunities for pricing and cost optimization.
- Strategic focus should be placed on **expanding high-performing markets, improving logistics efficiency, optimizing discount strategies, and strengthening customer retention** to drive sustainable business growth.

Business Insights

Analysis of the Supply Chain dataset provided several valuable business insights that can help management improve operational efficiency, profitability, and customer satisfaction.

1. Fishing Category Generated the Highest Revenue and Profit Among all product categories, Fishing consistently outperformed others by generating the highest sales revenue and overall profit. This indicates strong customer demand, high product acceptance, and better profit margins compared to other categories. Since this category contributes significantly to total business revenue, maintaining adequate inventory levels, improving supplier relationships, and introducing premium product variants can further increase profitability.

2. Consumer Segment Was the Largest Revenue Contributor The Consumer customer segment accounted for the largest proportion of total sales, contributing more than half of the company's overall revenue. This demonstrates that individual customers form the core customer base of the organization. Although Corporate and Home Office customers generated valuable revenue, the Consumer segment remains the primary driver of business growth. Customer retention programs, personalized marketing campaigns, and loyalty rewards should therefore focus heavily on this segment.

3. Western Europe Was the Best Performing Region Regional analysis revealed that Western Europe generated the highest sales among all geographical regions. Strong customer demand, consistent order volume, and higher purchasing power contributed to this excellent performance. The company should continue expanding its presence in this market by improving logistics infrastructure, increasing product availability, and investing in targeted regional marketing campaigns.

5. Seasonal Sales Decline During November and December The monthly sales trend indicated relatively stable business performance throughout most of the year. However, a noticeable decline in sales occurred during November and December. This seasonal slowdown may be caused by changing customer demand, inventory shortages, increased competition, or reduced purchasing activity. Early promotional campaigns, holiday offers, inventory planning, and targeted advertising can help reduce this decline in future years.

6. High Discounts Reduced Product Profitability Several products recorded strong sales volume but generated comparatively lower profit because of excessive

discounting. While discounts helped increase sales quantity, they significantly reduced profit margins. This highlights the importance of balancing promotional strategies with profitability objectives. Management should regularly review discount policies to ensure sustainable business growth.

7. The dashboard showed that a relatively small number of top-performing products contributed a significant portion of total company revenue and profit. These products should receive priority in procurement, warehouse allocation, and marketing investments to maximize future business performance.

8. Customer Spending Patterns Differ Across Segments Although Consumer customers generated the highest overall revenue, the Home Office segment recorded the highest average order value. This indicates that different customer segments have different purchasing behaviors and should be targeted with customized pricing strategies and promotional offers.

Recommendations

Based on the analytical findings, the following strategic recommendations are proposed to improve operational performance and maximize profitability.

1.Inventory Optimization

Increase inventory levels for high-demand categories such as Fishing, Cleats, and Camping & Hiking to prevent stock shortages and improve customer satisfaction. Demand forecasting should be used to optimize inventory planning and reduce stock-out risks.

2.Pricing and Discount Strategy

Review the current discount policy for products with low profit margins. Discounts should be offered selectively rather than uniformly to protect overall profitability while maintaining competitive pricing.

3.Improve Sales of Low-Performing Products

Introduce promotional campaigns, bundled offers, and targeted digital marketing strategies for products and categories experiencing weak sales performance. Poorly performing products should be continuously monitored to determine whether they should be improved or discontinued.

4.Regional Expansion Strategy

Continue investing in high-performing markets such as Western Europe while investigating the reasons behind weaker regional performance. Localized marketing campaigns and improved logistics can help increase sales in underperforming regions.

5.Customer Relationship Management

Develop customer loyalty programs for high-value customers, particularly within the Consumer segment. Personalized product recommendations and reward programs can improve customer retention and increase repeat purchases.

6.Logistics Improvement

Maintain the efficiency of Standard Shipping while reducing late deliveries through route optimization, warehouse coordination, and supplier performance monitoring. Faster and more reliable delivery directly improves customer satisfaction.

7.Data-Driven Decision Making

Business managers should regularly monitor KPIs through interactive dashboards to identify changing sales trends, customer behavior, and operational bottlenecks. Real-time monitoring enables faster and more informed strategic decisions.

Challenges

Several technical and business challenges were encountered during the development of this end-to-end Supply Chain Analytics project.

Large Dataset Processing

The dataset contained approximately 180,000 records, making data processing and transformation computationally intensive. Efficient data cleaning and optimized loading techniques were required to maintain dashboard performance.

Data Cleaning and Transformation

Several columns contained inconsistent values, missing records, duplicate entries, and different date formats. Power Query was extensively used to clean, standardize, and transform the raw data into a structured analytical dataset.

Date Format Conversion

The original dataset contained date values stored as text, which created challenges during PostgreSQL import and SQL analysis. Appropriate conversion functions and data transformation techniques were applied to convert text values into valid date formats.

Database Design

Designing an optimized Star Schema required separating the raw dataset into Fact and Dimension tables while maintaining relationships and referential integrity. Proper normalization improved SQL query performance and Power BI model efficiency.

SQL Query Development

More than 45 SQL queries were written to calculate KPIs, perform aggregation, identify trends, analyze customer behavior, and evaluate business performance. Query optimization techniques were applied to improve execution speed.

Dashboard Design

Creating multiple professional dashboards while maintaining a consistent color theme, navigation system, responsive layout, and business-friendly visualizations required several iterations and design improvements.

KPI Development

Developing meaningful KPIs required careful business understanding to ensure that management could easily interpret sales performance, customer behavior, profitability, logistics efficiency, and operational effectiveness.

Learning Outcomes

This project significantly enhanced both technical expertise and business analytical skills. Through this project, practical knowledge of the complete ETL (Extract, Transform, Load) process was gained, including data extraction, cleaning, transformation, validation, and loading into PostgreSQL. Hands-on experience was developed in designing relational databases and implementing dimensional modelling using Fact and Dimension tables within a Star Schema architecture. Advanced SQL skills were strengthened through writing complex analytical queries involving joins, aggregations, filtering, grouping, Common Table Expressions (CTEs), window functions, ranking functions, and KPI calculations. Practical experience in Power BI improved substantially by creating interactive dashboards, designing DAX measures, building KPI cards, implementing slicers, filters, bookmarks, drill-through analysis, and dynamic visualizations. Business understanding was enhanced by analyzing customer purchasing behaviour, regional sales performance, product profitability, logistics efficiency, and seasonal sales patterns. The project also improved problem-solving abilities, dashboard design principles, business storytelling techniques, and data visualization best practices that are commonly used in industry. Finally, this project provided valuable real-world experience in developing a complete end-to-end Business Intelligence solution from raw data collection to executive-level decision support.

Conclusion

The Supply Chain Executive Dashboard successfully converts raw operational data into strategic business intelligence. The solution demonstrates an end-to-end analytics workflow that supports data-driven decision-making and can be extended with forecasting, inventory optimization and machine learning in the future. The Supply Chain Executive Dashboard successfully transforms large volumes of raw operational data into meaningful business intelligence that supports strategic decision-making across multiple business functions. By integrating Excel, PostgreSQL, SQL, Power Query, Star Schema data modelling, DAX calculations, and Power BI visualizations, the project demonstrates a complete end-to-end Business Intelligence workflow aligned with industry best practices. The dashboards provide comprehensive visibility into sales performance, customer behaviour, product profitability, logistics operations, regional performance, and low-performing business areas through interactive reports and well-defined KPIs. Business users can quickly identify growth opportunities, operational inefficiencies, seasonal trends, and profitability issues without manually analyzing large datasets. The project highlights the importance of data-driven decision-making by converting complex business information into clear, actionable insights that support executives, managers, and operational teams. The recommendations generated from the analysis can help improve inventory management, optimize pricing strategies, strengthen customer relationships, enhance logistics efficiency, and increase overall business profitability. Overall, this project demonstrates strong practical skills in Business Intelligence, Data Analytics, Database Management, SQL, ETL Processes, Data Modelling, Dashboard Development, and Business Storytelling. It serves as a comprehensive portfolio project that reflects industry-standard analytical practices and can be further extended by incorporating predictive analytics, demand forecasting, inventory optimization, real-time data integration, cloud-based data warehousing, and machine learning models to support future business growth.